

JEE ADVANCED CUMULATIVE TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

Sec: JR.IIT_*CO-SC(MODEL-A)

Time: 3 Hrs

Date: 04-02-24

Max. Marks: 186

04-02-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_CAT-23_SYLLABUS

PHYSICS:

(100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 03-02-24

CHEMISTRY:

(100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 03-02-24

MATHEMATICS: (100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 03-02-24

THE NARAYANA GROUP

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

MATHEMATICS

Max. Marks: 62

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

1. The number of values of $x \in [0, 2\pi]$ for which

$$\left| \sqrt{2\sin^4 x + 18\cos^2 x} - \sqrt{2\cos^4 x + 18\sin^2 x} \right| = 1 \text{ is}$$

- A) 2 B) 6 C) 4 D) 8

2. Let $\{a_k\}$ and $\{b_k\}, k \in \mathbb{N}$ be two G.P.s with common ratio r_1 and r_2 respectively such that

$$a_1 = b_1 = 4 \text{ and } r_1 < r_2. \text{ Let } c_k = a_k + b_k, k \in \mathbb{N}. \text{ If } c_2 = 5 \text{ and } c_3 = \frac{13}{4} \text{ then } \sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$$

is equal to

- A) 3 B) 6 C) 9 D) 12

3. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4$, for all $x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \log_e(g(x))$, for all $x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is

- A) 1 B) 2 C) 3 D) 0

4. A line 'L' passing through origin is perpendicular to the lines

$$L_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$$

$$L_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, (s, t \in \mathbb{R})$$

If the co-ordinates of the point in the first octant on 'L₂' at a distance of $\sqrt{17}$ from the point of intersection of 'L' and 'L₁' are (a, b, c) then $18(a+b+c)$ is equal to _____

- A) 11 B) 22 C) 44 D) 33

5. Let $f(x) = \begin{cases} [x] & 0 \leq \{x\} < \frac{1}{2} \\ |x| & \frac{1}{2} \leq \{x\} < 1 \end{cases}, \forall x \in \left[-\frac{7}{2}, \frac{7}{2}\right]$. If L is number of points of discontinuity of

$f(x)$ on $\left[-\frac{7}{2}, \frac{7}{2}\right]$ and M is the number of points of non-differentiability of the function $f(x)$ on $\left(-\frac{7}{2}, \frac{7}{2}\right)$ then the value of $(L + M)$ is

[Note: $[y]$ and $\{y\}$ denotes greatest integer function less than or equal to y and fractional part of y respectively

- A) 9 B) 7 C) 20 D) 21

6. Let $S_n = \sum_{r=0}^{n-1} \cos^{-1} \left(\frac{n^2 + r^2 + r}{\sqrt{n^4 + r^4 + 2r^3 + 2n^2r^2 + 2n^2r + n^2 + r^2}} \right)$, then the value of S_{100} is

- A) $\frac{\pi}{12}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{6}$ D) $\frac{\pi}{4}$

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

7. Let $a_k = \frac{k}{(k-1)^{4/3} + k^{4/3} + (k+1)^{4/3}}$ and $S_n = \sum_{k=1}^n a_k$, then

- A) $S_{26} > \frac{17}{4}$ B) $S_{26} < \frac{17}{4}$ C) $S_{999} < 50$ D) $S_{999} > 50$

8. If A and B are two non singular matrices of order 3 such that $2A + 3BB^T = I$ and $B^{-1} = A^T$, then $(\text{Tr}(P)) = \text{trace of matrix P}$

- A) $|B^T - 2B + 3B^3 + 3BA| = 1$ B) $\text{Tr}(A^{-1} + I - AB - 3B^3) = 6$
C) $|A^{-1} - 3B^3| = 27$ D) $|A^{-1} - 3B^3| = 8$

9. Let $f(x) = \frac{1 - \cos\{x\}}{(x^4 + ax^3 + bx^2 + cx)^2}$. If $l = \lim_{x \rightarrow 1^+} f(x)$, $m = \lim_{x \rightarrow 2^+} f(x)$ and $n = \lim_{x \rightarrow 3^+} f(x)$, where l , m

and n are non-zero finite then: ($\{y\}$ denotes fractional part of y)

- A) $\lim_{x \rightarrow 0^+} f(x) = \frac{1}{36}$ B) $a + b + c = -1$
C) $l + m + n = \frac{19}{72}$ D) $\lim_{x \rightarrow 4^+} f(x) = 0$

10. Let $P(x)$ be a polynomial of **degree n** and $f(x) = \begin{cases} P\left(\frac{1}{x^3}\right) \cdot e^{\frac{-1}{x^4}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then
- A) $f(x)$ is discontinuous at $x = 0$ B) $f(x)$ is continuous at $x = 0$
- C) $f(x)$ is non differentiable at $x = 0$ D) $f'(0) = \lim_{x \rightarrow 0} f(x)$
11. Normals of parabola $y^2 = 4x$ at P and Q meets at $R(x_2, 0)$ and tangent at P and Q meets at $T(x_1, 0)$. Which of the following is/are **correct**?
- A) If $x_2 = 3$, then area of quadrilateral PTQR is 9
- B) If length of tangent PT is $4\sqrt{5}$, then $x_2 = 6$
- C) If length of tangent PT is $4\sqrt{5}$, then $x_2 = 7$
- D) If $x_2 = 4$ and area of circle circumscribing ΔPQR is $k\pi$, then k is equal to 9
12. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a non-zero vector perpendicular to $\vec{x}$ and $\vec{y} \times \vec{z}$ and $\vec{b}$ is non zero vector perpendicular to $\vec{y}$ and $\vec{z} \times \vec{x}$, then
- A) $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$ B) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$
- C) $\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$ D) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{z} - \vec{y})$
13. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ **be two functions** and $g \circ f : A \rightarrow C$ is defined. Then which of the following statement(s) is/are **incorrect**?
- A) If $g \circ f$ is onto then f must be onto
- B) If f is into and g is onto then $g \circ f$ must be onto function
- C) If $g \circ f$ is one-one then g is not necessarily one-one
- D) If f is injective and g is subjective then $g \circ f$ must be bijective mapping

14. A variable line 'L' is drawn through O (0, 0) to meet the lines $L_1 : y - x - 10 = 0$ and $L_2 : y - x - 20 = 0$ at points A and B respectively. A point P is taken on line 'L'.

- A) If $\frac{2}{OP} = \frac{1}{OA} + \frac{1}{OB}$, then locus of P is $3y - 3x = 40$
- B) If $OP^2 = (OA)(OB)$, then locus of P is $(y - x)^2 = 200$
- C) If $\frac{1}{OP^2} = \frac{1}{(OA)^2} + \frac{1}{(OB)^2}$, the locus of P is $(y - x)^2 = 80$
- D) If $OP^2 = (OA)(OB)$, then locus of P is $(y - x)^2 = 80$

SECTION-III
(Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 15 and 16

Let S_1 and S_2 be two fixed externally tangent circles with radius 2 and 3 respectively. Let S_3 be a variable circle internally tangent to both S_1 and S_2 at points A and B respectively. The tangents to S_3 at A and B meet at T, and $TA = 4$.

15. The radius of circle S_3 is equal to:

A) 2 B) 4 C) 6 D) 8

16. The area of circle circumscribing ΔTAB is

A) 10π B) 20π C) 40π D) 80π

Paragraph for Questions 17 and 18

Let PQ be a focal chord of the parabola $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a, a > 0$

17. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan \theta =$

A) $\frac{2}{3}\sqrt{7}$ B) $\frac{-2}{3}\sqrt{7}$ C) $\frac{2}{3}\sqrt{5}$ D) $\frac{-2}{3}\sqrt{5}$

18. Length of chord PQ is :

A) $7a$ B) $5a$ C) $2a$ D) $3a$

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

19. Two spherical stars A and B have densities ρ_A and ρ_B , respectively. Their radius R_A and R_B are related as $5R_A^3 = 2R_B^3$ and their masses M_A and M_B are related by $M_B = 2M_A$. Due to an interaction process, star A loses some of its mass, so that its radius is halved, while its spherical shape is retained, and its density remains ρ_A . The entire mass lost by A is deposited as a thick spherical shell on B with the density of the shell being ρ_A . If v_A and v_B are the escape velocities from A and B after the interaction process, the ratio $\frac{v_B}{v_A} = \sqrt{n}$.

The value of n is _____

- A) $\frac{25}{3}$ B) $\frac{23}{3}$ C) $\frac{31}{4}$ D) $\frac{35}{9}$
20. A small block having charge Q placed on a frictionless horizontal table is attached to one end of a spring of force constant k. The other end of the spring is attached to a fixed support and the spring stays horizontally in relaxed state. Another small block carrying a negative charge $-q$ is brought very slowly from a great distance towards the former block along the line coinciding with the axis of the spring. Find extension in the spring when both the blocks touch each other.

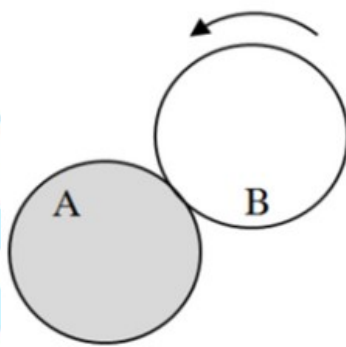
- A) $\frac{5}{2} \left(\frac{Qq}{2\pi\epsilon_0 k} \right)^{\frac{1}{3}}$ B) $\frac{3}{2} \left(\frac{Qq}{2\pi\epsilon_0 k} \right)^{\frac{1}{3}}$ C) $\frac{3}{4} \left(\frac{Qq}{2\pi\epsilon_0 k} \right)^{\frac{1}{3}}$ D) $\frac{5}{3} \left(\frac{Qq}{2\pi\epsilon_0 k} \right)^{\frac{1}{3}}$

21. At time $t = 0$, a disk of diameter 4m starts to roll without slipping on a horizontal plane with an angular acceleration of $\alpha = \frac{2}{3} \text{ rad s}^{-2}$. A small stone is stuck to the disk. At $t = 0$, it is at the contact point of the disk and the plane. Later, at time $t = \sqrt{\pi} \text{ s}$, the stone detaches itself and flies off tangentially from the disk. The maximum height (in m) reached by the stone measured from the plane is $\frac{1}{2} + \frac{x}{10}$.

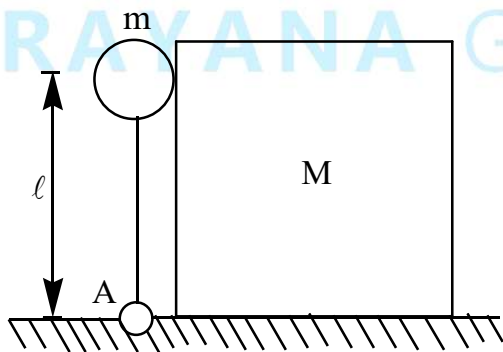
What is the value x ?

- A) $\pi + 4$ B) $(2\pi/3) + 5$ C) $(3\pi/4) + 9/2$ D) $(4\pi/5) + 11/3$

22. A flat surface of a thin uniform disk A of radius R is glued to a horizontal table. Another thin uniform ring B of mass M and with the same radius R rolls without slipping on the circumference of A, as shown in the figure. A flat surface of B also lies on the plane of the table. The centre of mass of B has fixed angular speed ω about the vertical axis passing through the centre of A. The angular momentum of B is $nM\omega R^2$ with respect to the centre of A. Which of the following is the value of n?



- A) 2 B) 6 C) $\frac{7}{2}$ D) $\frac{9}{2}$
23. A weightless rod of length ℓ with a small load of mass m at the upper end is hinged at point A as shown in the figure. Initially the rod is vertical and m touches another block of mass M. A light jerk sets the system in motion. For a given mass ratio $\frac{M}{m} = x$, when the rod forms an angle $\alpha = \frac{\pi}{6}$ with the horizontal, the body M separates from the mass m. Friction should be neglected everywhere. What is the speed of the body mass M at the time of separation?(all the bodies move in the same vertical plane)



- A) $\sqrt{\frac{g\ell}{2}}$ B) $\sqrt{\frac{g\ell}{8}}$ C) $\sqrt{\frac{g\ell}{3}}$ D) $\sqrt{\frac{g\ell}{16}}$

24. A medium having dielectric constant $K > 1$ fills the space between the plates of a parallel plate capacitor. The plates have large area, and the distance between them is d . The capacitor is connected to a battery of voltage V . As shown in figure (a). Now, both the plates are moved by a distance of $\frac{d}{2}$ from their original positions, as shown in Figure (b).

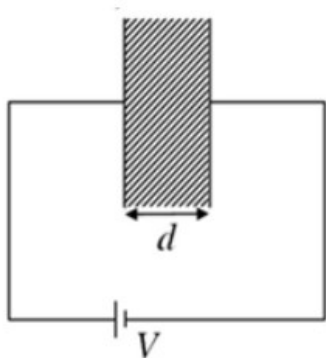


Figure (a)

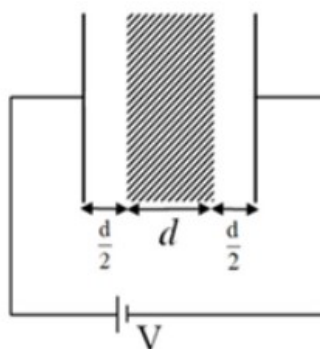


Figure (b)

In the process of going from the configuration depicted in Figure (a) to that in Figure (b), which of the following statement is correct?

- A) The electric field inside the dielectric material is reduced by a factor of K .
- B) Final capacitance is $\frac{1}{K+1}$ times the initial capacitance
- C) The voltage between the capacitor plates is increased by a factor of $(K+1)$
- D) The work done in the process DOES NOT depend on the presence of the dielectric material.

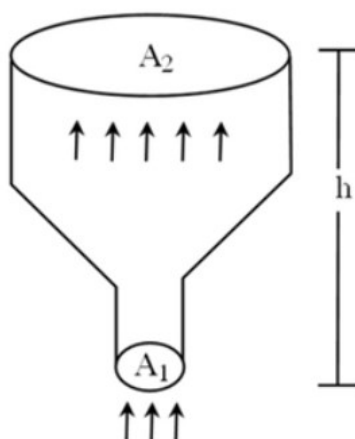
SECTION-II (Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

25. An ideal gas of density $\rho = 0.2 \text{ kg m}^{-3}$ enters a chimney of height h at the rate of $\alpha = 0.8 \text{ kg s}^{-1}$ from its lower end, and escapes through the upper end as shown in the figure. The cross-sectional area of the lower end is $A_1 = 0.1 \text{ m}^2$ and the upper end is $A_2 = 0.4 \text{ m}^2$. The pressure and the temperature of the gas at the lower end are 600 Pa and 300 K, respectively, while its temperature at the upper end is 150 K. The chimney is heat

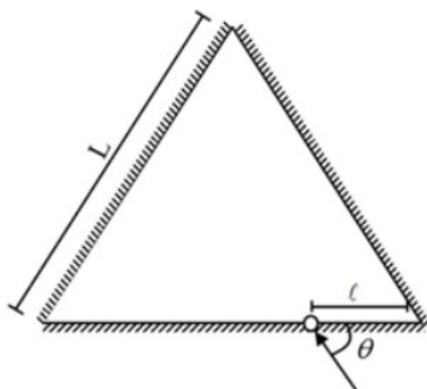
insulated so that the gas undergoes adiabatic expansion. Take $g = 10 \text{ ms}^{-2}$ and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure.



Which of the following statement (s) is(are) correct?

- A) The pressure of the gas at the upper end of the chimney is 150 Pa.
- B) The velocity of the gas at the lower end of the chimney is 40 ms^{-1} and at the upper end is 20 ms^{-1}
- C) The height of the chimney is 360 m.
- D) The density of the gas at the upper end is 0.5 kg m^{-3}

26. Three plane mirrors form an equilateral triangle with each side of length L . There is a small hole at a distance $\ell > 0$ from one of the corners as shown in the figure. A ray of light is passed through the hole at an angle θ and can only come out through the same hole. The cross section of the mirror configuration and the ray of light lie on the same plane.



Which of the following statement(s) is(are) correct?

- A) The ray of light will come out for $\theta = 30^\circ$, for $0 < \ell < L$.
- B) There is an angle for $\ell = \frac{L}{2}$ at which the ray of light will come out after two reflections
- C) The ray of light will NEVER come out for $\theta = 60^\circ$, and $\ell = \frac{L}{3}$
- D) The ray of light will come out for $\theta = 60^\circ$, and $0 < \ell < \frac{L}{2}$ after five reflections.

27. A disk of radius R with uniform positive charge density σ is placed on the xy plane with its center at the origin. A particle of positive charge q is placed initially at rest at a point on the z axis with $z = z_0$ and $z_0 > 0$. In addition to the Coulomb force, the particle experiences a vertical force $\vec{F} = -c\hat{k}$ with $c > 0$. Let $\beta = \frac{2c\epsilon_0}{q\sigma}$. Which of the following statement(s) is (are) correct?

- A) For $\beta = \frac{1}{4}$ and $z_0 = \frac{25}{7}R$, the particle reaches the origin.
- B) For $\beta = \frac{1}{4}$ and $z_0 = \frac{R}{\sqrt{3}}$, the particle returns back to $z = z_0$.
- C) For $\beta = \frac{1}{4}$ and $z_0 = \frac{R}{\sqrt{3}}$, the particle enters the negative z -axis
- D) For $\beta > 1$ and $z_0 > 0$, the particle always reaches the origin.

28. A bubble has surface tension S . The ideal gas inside the bubble has ratio of specific heats $\gamma = \frac{5}{3}$. The bubble is exposed to the atmosphere and it always retains its spherical shape. When the atmospheric pressure is P_{a1} , the radius of the bubble is found to be r_1 and the temperature of the enclosed gas is T_1 . When the atmospheric pressure is P_{a2} , the radius of the bubble and the temperature of the enclosed gas are r_2 and T_2 , respectively. Which of the following statement(s) is (are) correct?

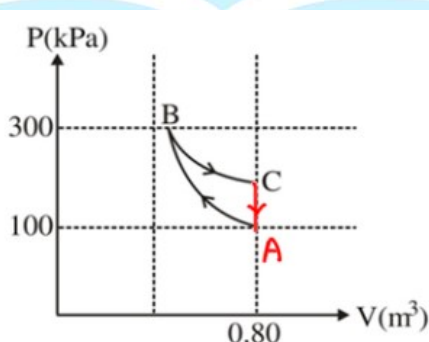
A) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{r_1}{r_2}\right)^5 = \frac{P_{a2} + \frac{2S}{r_2}}{P_{a1} + \frac{2S}{r_1}}$

B) If the surface of the bubble is a perfect heat insulator, then the total internal energy of the bubble including its surface energy does not change with the external atmospheric pressure.

C) If the surface of the bubble is a perfect heat conductor and the change in atmospheric temperature is negligible, then $\left(\frac{r_1}{r_2}\right)^3 = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$

D) If the surface of the bubble is perfect heat insulator, then $\left(\frac{T_2}{T_1}\right)^{\frac{5}{2}} = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$.

29. In the given P – V diagram, a monoatomic gas $\left(\gamma = \frac{5}{3}\right)$ is first compressed adiabatically from state A to state B. then it expands isothermally from state B to state C and then brought back to state A isochorically. Which of the following statement(s) is (are) correct? [Given : $\left(\frac{1}{3}\right)^{0.6} \approx 0.5, \ln 2 \approx 0.7]$



- A) The magnitude of the total work done in the process $A \rightarrow B \rightarrow C$ is 24 kJ.
- B) The magnitude of the work done in the process $B \rightarrow C$ is 84 kJ
- C) The magnitude of the work done in the process $A \rightarrow B$ is 60 kJ
- D) The magnitude of the work done in the process $C \rightarrow A$ is zero.

30. A thin rod of length 10 cm is placed along the axis of a concave mirror of focal length 30 cm in such a way that one end of the image coincides with one end of the object. The length of the image may be

- A) 7.5 cm B) 12 cm C) 15 cm D) 10 cm

31. A particle starting from rest at the highest point slides down the outside of a smooth vertical circular track of radius 0.3m. When it leaves the track its vertical fall is h and the linear velocity is v . The angle made by the radius at that position of the particle with the vertical is θ . Now consider the following observations : ($g = 10 \text{ m/s}^2$)

(I) $h = 0.1 \text{ m}$ and $\cos \theta = 2/3$

(II) $h = 0.2 \text{ m}$ and $\cos \theta = 1/3$

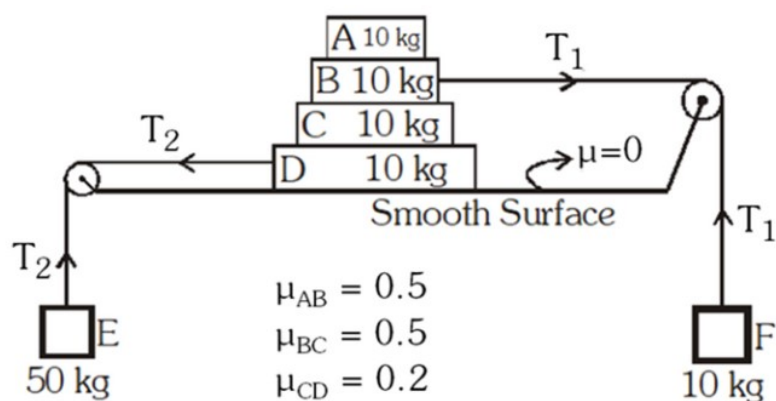
(III) $v = \sqrt{2} \text{ m/s}$

(IV) after leaving the circular track, the particle will describe a parabolic path

A) (I) and (III) both are correct B) Only (II) is incorrect

C) only (III) is correct D) (IV) is correct

32. Consider all pulleys and string to be ideal.



A) The value of T_2 is 200 N

B) The value of T_2 is $\frac{400}{3}$ N

C) The force of friction between B and C is 70 N

D) The force of friction between B and C is 90 N

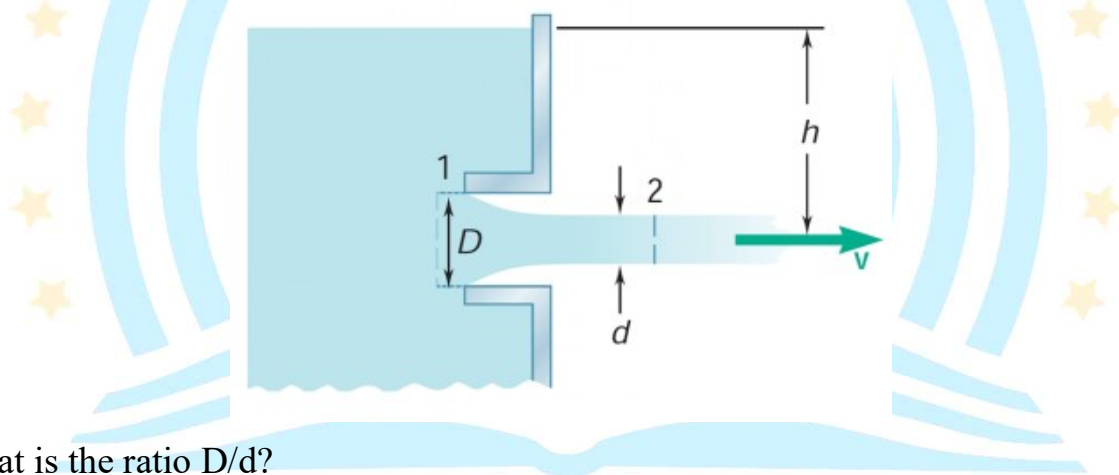
SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 33 and 34:

Usually in context of JEE Physics, (when an ideal fluid comes out of an orifice) we assume that when a jet issues from a small orifice, the velocity of particles is perpendicular to the orifice area, however this is not true because in general the fluid particles will also have a radial velocity and it will take some distance for the streamlines to actually become nearly parallel (see figure). In the adjacent figure, we may assume that near the walls of the vessel the pressure is hydrostatic (Though it is not so near the mouth of re-entrant orifice (an orifice whose pipe starts well inside the volume of the container as shown in the figure)). It turns out that by careful application of Bernoulli's equation and Momentum Equations, we can estimate the ratio of diameters of stream (d) where the streamlines become nearly parallel (section 2) to the diameter of the orifice (D). Here we have a rectangular tank which is filled to a height h above the small orifice of diameter D as shown. **Neglect atmospheric pressure everywhere.** Based on this passage now answer the following questions :



33. What is the ratio D/d ?

- A) $\sqrt{2}$ B) $\sqrt{3}$ C) $\frac{\sqrt{5}}{2}$ D) $\frac{\sqrt{7}}{3}$

34. What is the average horizontal component of velocity immediately after exiting the orifice (i.e. Flow rate / hole cross sectional area)

- A) $\sqrt{gh}$ B) $\frac{2}{3}\sqrt{gh}$ C) $\sqrt{\frac{gh}{2}}$ D) $\frac{2}{5}\sqrt{gh}$

Paragraph for Questions 35 and 36:

An isotropic point source on the ground sprays water droplets at speed v uniformly in directions which make an angle between 0 and $\frac{\pi}{4}$ radians with the vertical. Define the origin to be at the point source and consider the plane of the ground. Suppose that the wetness at a point on the ground that is a distance r away from the origin is proportional to the number of water droplets impinging the immediate neighbourhood of that point per unit area.

35. Determine the radial coordinate r of a point that is $\frac{2}{(\sqrt{6}-\sqrt{2})}$ times as wet as the point that is a distance $\frac{v^2}{2g}$ away from the origin.

(You may need to solve a cubic trigonometric equation by hit and trial using standard angles or factorization)

- A) $\sqrt{5} \frac{v^2}{4g}$ B) $\sqrt{3} \frac{v^2}{3g}$ C) $\sqrt{5} \frac{v^2}{3g}$ D) $\sqrt{3} \frac{v^2}{2g}$
36. What is the radial coordinate of wettest points?
- A) $\frac{v^2}{2g}$ B) $\frac{3v^2}{4g}$ C) $\frac{v^2}{g}$ D) $\frac{v^2}{\sqrt{2}g}$

THE NARAYANA GROUP

CHEMISTRY

Max. Marks: 62

SECTION-I

(Single Correct Answer Type)

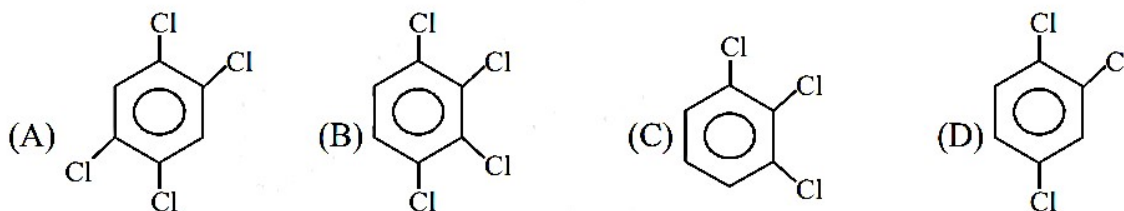
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

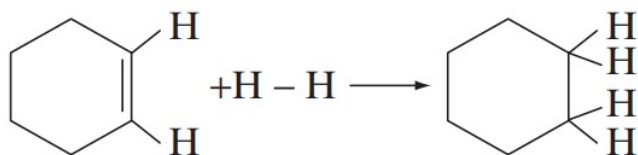
37. Which of the following molecule has two lone pairs on its central atom?

- A) SF_4 B) I_3^- C) XeF_5^- D) $XeOF_4$

38. Which of the given compound has highest dipole moment ?



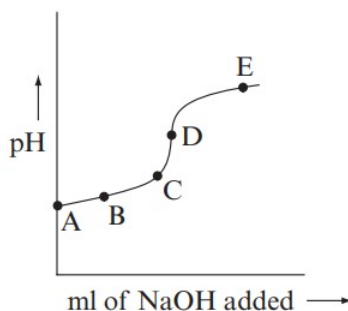
39.



The bond energies (in kJ/mol) at 25°C are C–C = 346, C–H = 413, H–H = 437, C = C; 611. From these data, the value of ΔH at 25°C for the above reaction is

- A) -289 kJ mol^{-1} B) -124 kJ mol^{-1} C) 124 kJ mol^{-1} D) 289 kJ mol^{-1}

40. The curve in the figure shows the variation of pH during the course of titration of a weak acid, HA with a strong base (NaOH). At which point in the titration curve is the concentration of the acid may be equal to that of its conjugate base?



- A) Point D B) Point E C) Point C D) Point B

41. The vapour pressure of water at room temperature is lowered by 5% on dissolving a non-volatile solute in it. The molality of the solution is
 A) 2.0 B) 1.4 C) 2.78 D) 2.92
42. For which orbit in He^+ ion, the circumference is $26.5 \text{ \AA}$?
 A) 2 B) 3 C) 4 D) 16

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

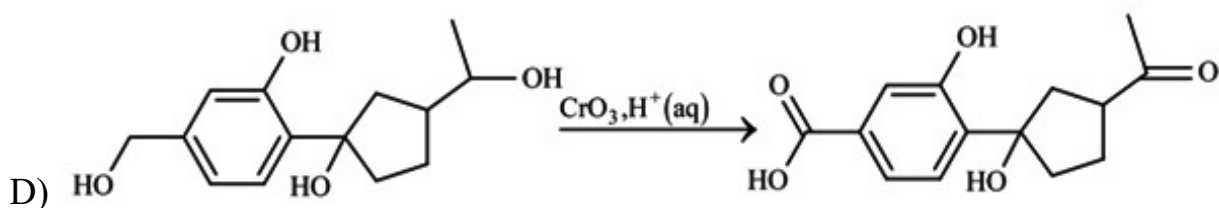
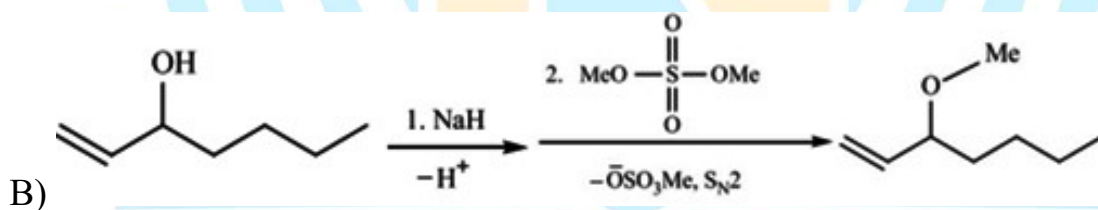
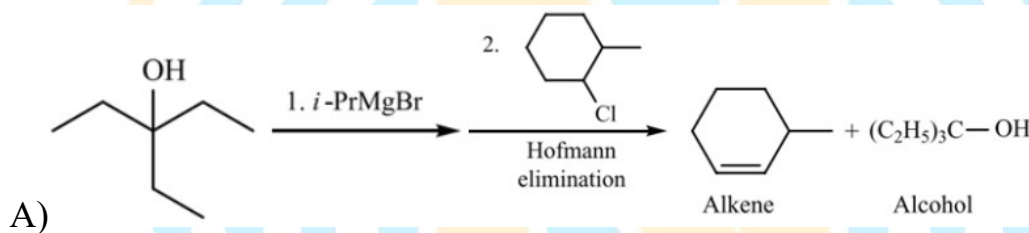
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

43. Which of the following overlaps of orbitals would lead to the formation of π -bond?

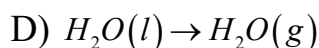
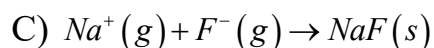
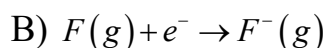
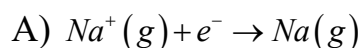
(Internuclear axis is z-axis)

- A) $p_x - p_x$ B) $d_{yz} - d_{yz}$ C) $d_{xy} - d_{xy}$ D) $d_{xz} - d_{xz}$

44. Which of the following are correct?



45. The enthalpy change for the following process would be expected to be a negative for



46. Benzene and naphthalene forms an ideal solution at room temperature. For this process, the true statement(s) is(are)

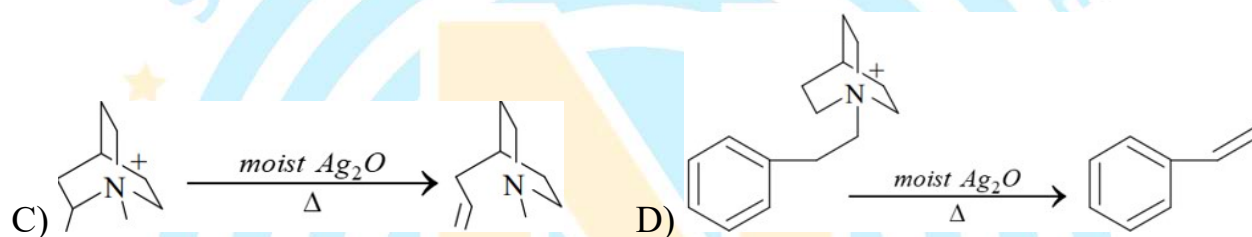
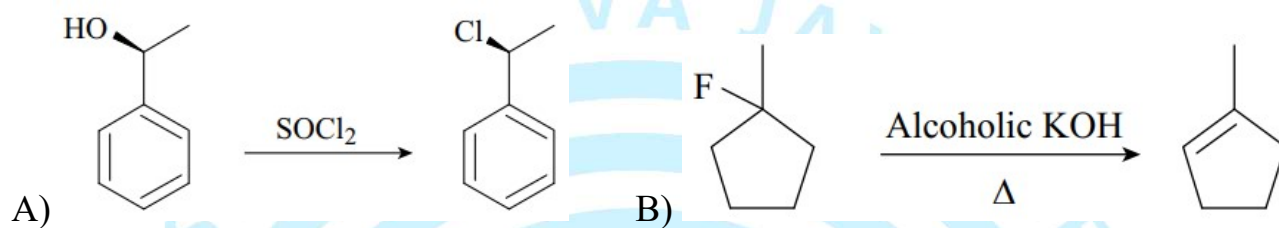
A) $\Delta G = +ve$

B) $\Delta S_{\text{system}} = +ve$

C) $\Delta S_{\text{surrounding}} = 0$

D) $\Delta H = 0$

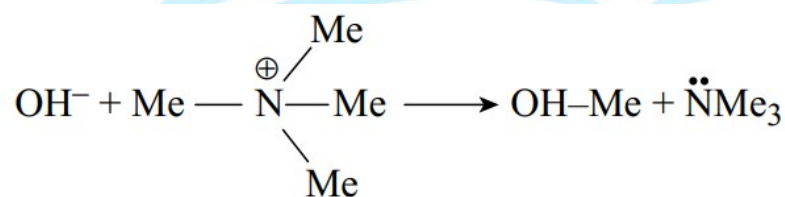
47. Identify reactions that correctly match with their products



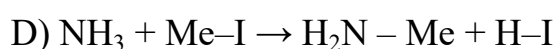
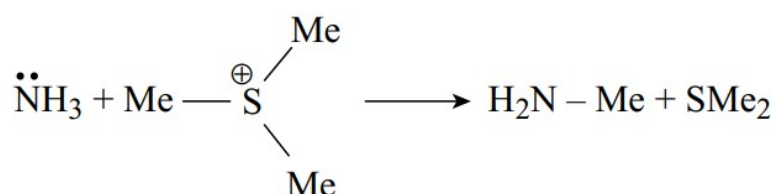
48. Among the following $\text{S}_{\text{N}}2$ reactions, which will become favourable by increasing the polarity of the medium?



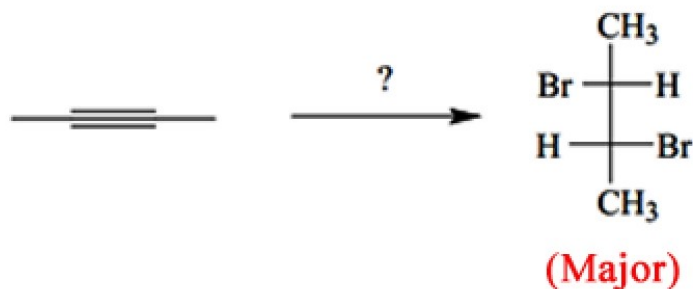
B)



C)

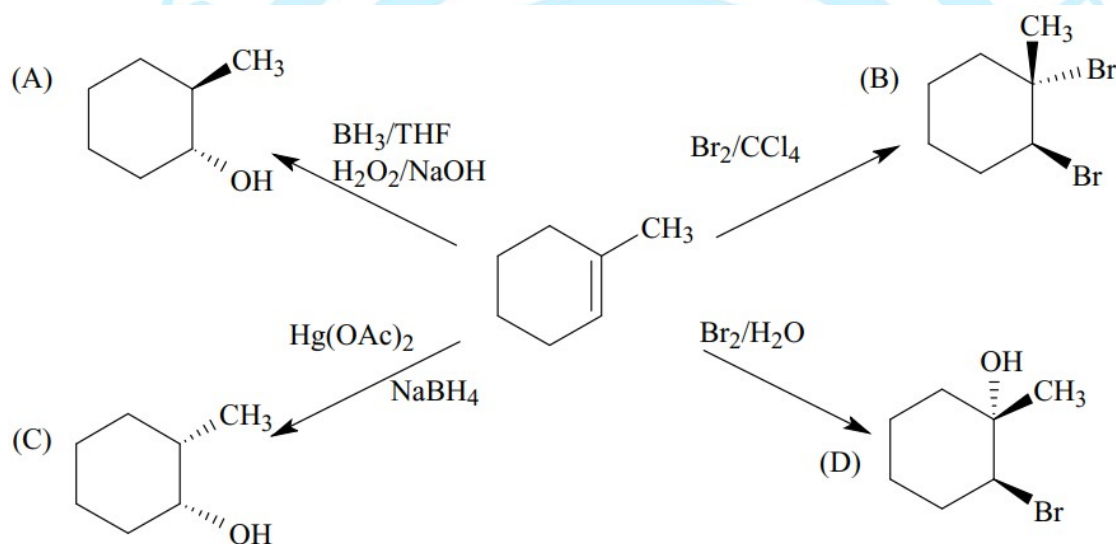


49.



- A) (i) Na/NH₃ (l) (ii) Br₂ /CHCl₃
 B) (i) H₂ /Pd/CaCO₃ /pyridine (ii) HBr
 C) (i) excess H₂ /Pd (ii) Br₂ ,uv light
 D) (i) H₂ /Pd/CaCO₃ /pyridine (ii) Br₂ /CHCl₃

50. Identify the correct reaction sequences (consider the given product as major product)



SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B),(C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 51 and 52:

The term carbide is generally applied to compounds in which carbon is bonded to the elements of lower or approximately same electronegativity. This definition excludes the compounds in which oxygen, sulphur, phosphorus nitrogen and halogens are united with carbon. Reactive metals (i.e. of group 1 and 2) form ionic carbides. They hydrolyse to liberate hydrocarbons. Most of them resembles with NaCl in crystal structure. Transition metals forms interstitial carbides.

51. Read the following statement regarding metal carbide.

- (I) All metal carbides are ionic
- (II) CaC_2 has one sigma and two pi bonds
- (III) Be_2C on hydrolysis forms amphoteric hydroxide
- (IV) SiC is 3-D covalent solid compound having sp^2 hybridised Si atoms.

Choose the correct options :

- A) Only II B) II and III C) II, III and IV D) II and IV

52. Which of the following statements is incorrect regarding carbides?

- A) Calcium carbide (CaC_2) reacts with water to produce acetylene gas.
- B) Tungsten carbide (WC) is commonly used in cutting tools and drill bits.
- C) Aluminum carbide (Al_4C_3) undergoes hydrolysis to produce propane gas.
- D) Silicon carbide (SiC) is a covalent compound with a high melting point and is commonly used as an abrasive.

Paragraph for Questions 53 and 54:

The specific rotation of $(2R, 3R) - (+) -$ tartaric acid is $+12.4^\circ(\text{H}_2\text{O})$. The optical rotation of a solution made up from 10 g of a mixture optically active tartaric acid in 20 ml of water was measured in a 10 cm cell. The observed rotation was -3.1° .

53. What is the specific rotation of the mixture?

- A) -3.1° B) -6.2° C) -0.62° D) $+12.4^\circ$

54. What is the percentage of $(2R, 3R) - (+) -$ tartaric acid in the solution?

- A) 12.5 % B) 50% C) 75% D) 25 %